

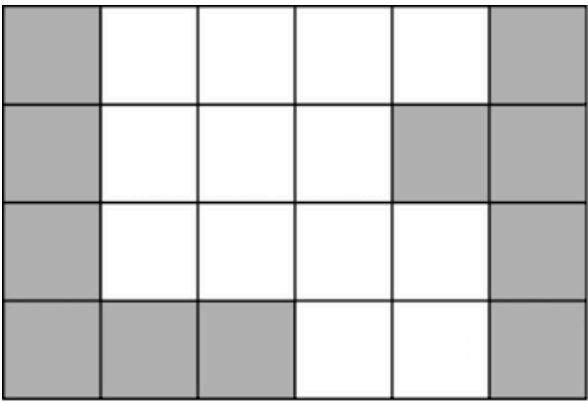
Problem A. Cahokia ruins

Time Limit 2000 ms
Mem Limit 262144 kB
OS Windows

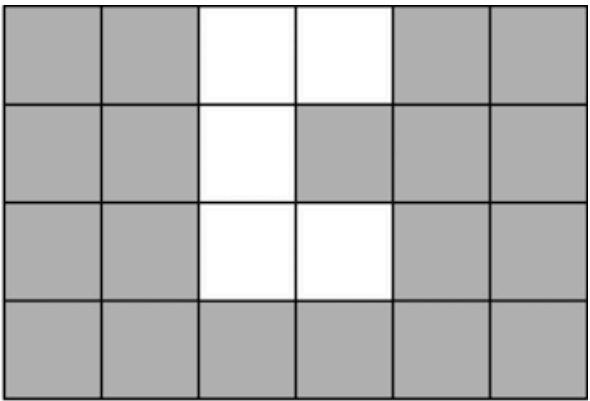
Cahokia is the area of an ancient civilization located in the state of Illinois. This place importance comes from the fact that it helps understand the history of North America. Cahokia was composed by 120 mounds, 80 of them still remain nowadays. One of these mounds is called Monks Mound, considered the biggest natural monument in all North America.

Recently, a group of archaeologist of the organization Extraordinary Mystery Investigators (IME, in their language) have discovered secret passages in the underground surroundings of Monks Mound. Because of the importance of this discovery, IME has called one of its famous members, Marcio "the indispensable" Himura, for help. After diving in Thai seas, Marcio is back in action and want to unfold the mysteries of Cahokia. The members of IME want to reach the deepest part of Monk Mound, but to do so, they have to traverse a special room.

This room can be represented by a 2D grid with dimensions $H \times W$ meters. The particularity of this room is that the instant somebody enters the room the east and west walls begin to move towards each other until they collide. Marcio suspects that this was made to keep safely treasures from intruders. Each wall is represented by a sequence of H integers $a_1, a_2, \dots, a_H$, where a_i is the width (in meters) of the wall at row i . Marcio knows that each wall moves with speed of 1 m/s . The following figure depicts (a) the initial configuration of the room given in the first sample test case, and (b) the collision of the walls after one second.



(a)



(b)

Marcio is interested in knowing, given an initial configuration of the walls, in how much time the walls collide after someone enter the room. This information is crucial for Marcio to determine an strategy to go through the room. He acknowledges your potential, so he ask for your help to solve this problem.

Input

In the first line we give two space-separated integers, H ($2 \leq H \leq 10^5$) and W ($4 \leq W \leq 10^5$). In the second line, there are H space-separated integers representing the east wall. In the third line, there are H space separated-integers representing the west wall. It is ensured that the walls don't overlap at any row.

Output

One number representing in how much time the two walls hit each other. Error of 10^{-9} is tolerated.

Examples

Input	Output
4 6 1 1 1 3 1 2 1 1	1

Input	Output
4 100 10 10 20 30 15 35 20 10	27.5

Problem B. Black Hills golden jewels

Time Limit 2000 ms

Mem Limit 262144 kB

OS Windows

In Rapid City are located the main producers of the *Black Hills gold jewelry*, a very popular product among tourists. The main characteristics of these jewels are the designs of leaves and grape clusters that are made with gold alloys in yellow, green and pink tones.

Planning a trip to Rapid City in 2017, Nay Suames decides that he wants to buy these famous jewels, but he doesn't have much money to spend. Searching for the lowest prices, he found a jewelry store that will have a sale in May, the same month he will travel to that city. Lucky guy!

The jewelry store intends to sell the jewels that were produced with small defects, in general, only perceivable by experts. The jewels are separated in N categories of defects. Even if the defect is imperceptible, the jewelry store can't sell the jewel at full price, so it gives a discount proportional to the defect.

Knowing the high demand for these jewels, the jewelry store created a system aiming to be fair and, at the same time, to prevent everybody from buying only the cheapest jewels. The clients have to make a queue and will be served one at a time. When their time comes, the client must choose two jewels from different categories. Furthermore, they can't choose the same combination of categories that was previously chosen by another client.

Nay asks for your help to find the minimum amount of money he needs, being the K -th client in the queue, so that it is guaranteed that he can buy two different jewels.

Input

The first line contains two integers, N and K , ($2 \leq N \leq 10^5$, $1 \leq K \leq \frac{N(N-1)}{2}$), representing the number of categories and the position of Nay in the queue, respectively. The next line contains N integers $a_1, a_2, \dots, a_N$ ($0 \leq a_i \leq 10^9$), corresponding to the prices of each category.

Output

Print a single integer, the minimum amount of money to guarantee that Nay can buy the jewels.

Examples

Input	Output
4 3 2 4 5 2	6

Input	Output
6 9 1 2 3 4 5 6	7

Problem C. Metal detector

Time Limit 500 ms

Mem Limit 262144 kB

OS Windows

Since the 2017 world finals will be in Rapid City, South Dakota, the MaratonIME's coach decides to visit the city to know the contest location.

The city is preparing to receive the best computer programmers in the world. To ensure the safety of the event, metal detectors are being installed at the entrance of the convention center where the contest will be held. To avoid long waits in the line, the security officers were instructed to send every person that activates the metal detector to the end of the line.

The coach noticed that one of the metal detectors is defective and it is always activated alternately. This means that the first person does not activate it and the next person does. The third person does not activate it and the next person does, and so on.

Given that there are N people in the line, the coach, that is the i -th in the line, wants to know when he will be able to go through the metal detector without activating it.

Input

The input starts with an integer T , the number of test cases.

Each test case consists of two integers, N and i , corresponding to the number of people in the line and the position of the coach in the line, respectively.

Limits

- $1 \leq T \leq 10^3$
- $1 \leq N \leq 10^9$
- $1 \leq i \leq N$

Output

For each test case, print a line with one integer j meaning that the coach will be the j -th person to go through the metal detector without activating it.

Examples

Input	Output
2 5 1 6 2	1 4

Problem D. The Declaration of Independence

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

In 1776, a Committee of Five was chosen to draft the Declaration of Independence of the USA, among them John Adams, Thomas Jefferson and Benjamin Franklin. From June 11 to July 5 they worked tirelessly to write such important document.

It wasn't written correctly in the first attempt, and many many changes had to be made. Since in that time there weren't such great and time-saving utilities like `git` and version control, they had a very simple but time consuming way to keep old versions of the document.

Suppose Thomas Jefferson wanted to add a line to a version i of the document. Instead of changing the actual document in version i , he would use a letter copying press, a machine just invented by James Watt just around that time, to copy the version i to a new sheet of paper, and then would modify this new sheet of paper. This paper would then be stored so it could be copied later. As each modification creates a new version of the document, the k th modification will create the version k of the document. You can assume the version 0 of the document is an empty piece of paper.

As everything was written in ink and the committee doesn't like to contradict itself, each modification could only add some lines to the end of the document, or erase some lines from the beginning of the document.

Your task is to simulate the making of the Declaration of Independence. Each sentence is represented as an integer number. You need to process the following queries:

- $E \ v \ x$ — Copy the document with version v , then add sentence x to the end of the copied document
- $D \ v$ — Copy the document with version v , then remove the first sentence of the copied document

Queries are numbered from one. The document with version 0 is empty.

Input

In the first line an integer Q , the number of queries. Each of the next Q lines has a query, in the format given in the statement.

Limits

- $1 \leq Q \leq 10^5$
- In the i th query it is guaranteed that $0 \leq v < i$
- For each query of type E, x will fit in a 32-bit signed integer
- For each query of type D, it is guaranteed the document will not be empty during the removal of the first sentence

Output

For each query of type D, print the sentence removed.

Examples

Input	Output
8 E 0 10 E 0 -10 D 2 D 1 E 2 5 D 5 D 6 D 2	-10 10 -10 5 -10

Problem E. Pop Divas

Time Limit 2000 ms

Mem Limit 262144 kB

OS Windows

Glauber "the drinker" Sokolov, after participating in two ICPC World Finals, decided to abandon competitive programming and pursue his dream of working with the pop divas. His first job was as a dancer in one of Britney Spears' concerts, then composing songs for Lana del Rey, and he worked with Rihanna, Shakira, and many others. As time passed he used his unmatched intelligence to rise to the top, and became a producer.

Each song has an attribute named quality, that depends on how catchy it is, how touching the lyrics are and how danceable is the beat. This value can be represented as a number in the range $[0, +\infty]$. When choosing the tracks for an album, the quality of an album is equal to the glauber mean of the qualities all the songs in that album, that is, if $a_1, \dots, a_B$ are the qualities of the songs chosen for an album, then the quality of the album is $\sqrt{a_1 \cdot a_2 \cdot \dots \cdot a_B}$.

The quality of an album is closely related to how much it will sell and how it will be portrayed on the media. Producers don't usually know the quality of their singers' songs, but Glauber is no simple producer.

Glauber manages both Katy Perry and Taylor Swift music, and both divas are about to release an album. Each singer has a collection of songs, and Glauber will choose one or more songs from that collection to create an album. As both singers are very good, Glauber knows the quality of all of their songs are integers not smaller than one.

To avoid creating any more bad blood between them, Glauber wants to choose albums with the same quality, so that the media won't compare them and rate one as better than the other. He forgot how to do that since he stopped programming, can you help him?

Input

The first line has two integers N and M , the size of the collection of songs from Taylor Swift and Katy Perry, respectively.

The second line has N integers $a_1, \dots, a_N$, the qualities of the songs from Taylor.

The third line has M integers $b_1, \dots, b_M$, the qualities of the songs from Katy.

Limits

- $1 \leq N, M \leq 16$
- $1 \leq a_i, b_j \leq 10^9$

Output

In the first line print "Y" if it is possible to choose such albums, and "N" otherwise.

If it is possible, print the sizes of the albums from Taylor and Katy in one line, and the quality of their songs in the next two lines, in the same format as in the input. The values for each album can be printed in any order. If there are multiple answers, you can print any of them.

Examples

Input	Output
2 3 2 15 6 5 7	Y 2 2 2 15 6 5

Input	Output
3 5 2 3 5 7 11 13 17 19	N

Problem F. Mount Rushmore and Birthdays

Time Limit 250 ms
Mem Limit 65536 kB
OS Windows

After going to the World Finals 2017, your team decided to stay in Rapid City for an extra week and visit some nearby attractions.

When visiting Mount Rushmore, a sculpture with the carved faces of presidents George Washington, Thomas Jefferson, Theodore Roosevelt and Abraham Lincoln, you wondered what was the chance that two of them had birthday in the same day. After a quick search on the internet, you find out that Lincoln was born on February 12 and Washington on February 11. Close enough!

The chance that in a group of 4 people 2 of them have the same birthdays is really quite low, about 1.6%. But your teammate tells you that in a group of 23 random people, the chance that some pair of them will have the same birthday is greater than 50%!

This is interesting, but if the year had a different number of days, how would this number change?

Given N , the number of days in a year, answer what is the minimum number of people needed in a room so that the chance that a pair of them has the same birthday is greater than 50%. You can assume each day has the same chance of being someone's birthday.

Input

The input has a single integer N , the number of days in a year.

Limits

- $1 \leq N \leq 1000$

Output

Print a single integer, the answer to the problem.

Examples

Input	Output
365	23

Input	Output
666	31

Problem G. School Dance

Time Limit 1000 ms

Mem Limit 524288 kB

There are n boys and m girls in a school. Next week a school dance will be organized. A dance pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Input

The first input line has three integers n , m and k : the number of boys, girls, and potential pairs. The boys are numbered $1, 2, \dots, n$, and the girls are numbered $1, 2, \dots, m$.

After this, there are k lines describing the potential pairs. Each line has two integers a and b : boy a and girl b are willing to dance together.

Output

First print one integer r : the maximum number of dance pairs. After this, print r lines describing the pairs. You can print any valid solution.

Constraints

- $1 \leq n, m \leq 500$
- $1 \leq k \leq 1000$
- $1 \leq a \leq n$
- $1 \leq b \leq m$

Example

Input	Output
3 2 4 1 1 1 2 2 1 3 1	2 1 2 3 1

Problem H. D-query

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

[English](#)
[Vietnamese](#)

Given a sequence of n numbers $a_1, a_2 \dots a_n$ and a number of d -queries. A d -query is a pair (i, j) ($1 \leq i \leq j \leq n$). For each d -query (i, j) , you have to return the number of distinct elements in the subsequence $a_i, a_{i+1} \dots a_j$.

Input

- Line 1: n ($1 \leq n \leq 30000$).
- Line 2: n numbers $a_1, a_2 \dots a_n$ ($1 \leq a_i \leq 10^6$).
- Line 3: q ($1 \leq q \leq 200000$), the number of d -queries.
- In the next q lines, each line contains 2 numbers i, j representing a d -query ($1 \leq i \leq j \leq n$).

Output

- For each d -query (i, j) , print the number of distinct elements in the subsequence $a_i, a_{i+1} \dots a_j$ in a single line.

Example

Input	Output
5 1 1 2 1 3 3 1 5 2 4 3 5	3 2 3

Problem I. Planets and Kingdoms

Time Limit 1000 ms

Mem Limit 524288 kB

A game has n planets, connected by m teleporters. Two planets a and b belong to the same kingdom exactly when there is a route both from a to b and from b to a . Your task is to determine for each planet its kingdom.

Input

The first input line has two integers n and m : the number of planets and teleporters. The planets are numbered $1, 2, \dots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b : you can travel from planet a to planet b through a teleporter.

Output

First print an integer k : the number of kingdoms. After this, print for each planet a kingdom label between 1 and k . You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 6 1 2 2 3 3 1 3 4 4 5 5 4	2 1 1 1 2 2

Problem J. Mail Delivery

Time Limit 1000 ms
Mem Limit 524288 kB

Your task is to deliver mail to the inhabitants of a city. For this reason, you want to find a route whose starting and ending point are the post office, and that goes through every street exactly once.

Input

The first input line has two integers n and m : the number of crossings and streets. The crossings are numbered $1, 2, \dots, n$, and the post office is located at crossing 1.

After that, there are m lines describing the streets. Each line has two integers a and b : there is a street between crossings a and b . All streets are two-way streets.

Every street is between two different crossings, and there is at most one street between two crossings.

Output

Print all the crossings on the route in the order you will visit them. You can print any valid solution.

If there are no solutions, print "IMPOSSIBLE".

Constraints

$$2 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \cdot 10^5$$

$$1 \leq a, b \leq n$$

Example

Input	Output
6 8 1 2 1 3 2 3 2 4 2 6 3 5 3 6 4 5	1 2 6 3 2 4 5 3 1

Problem K. Police Query

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

To help capture criminals on the run, the police are introducing a new computer system. The area covered by the police contains N cities and E bidirectional roads connecting them. The cities are labelled 1 to N . The police often want to catch criminals trying to get from one city to another. Inspectors, looking at a map, try to determine where to set up barricades and roadblocks.

The new computer system should answer the following two types of queries:

1. Consider two cities A and B , and a road connecting cities G_1 and G_2 . Can the criminals get from city A to city B if that one road is blocked and the criminals can't use it?
2. Consider three cities A , B and C . Can the criminals get from city A to city B if the entire city C is cut off and the criminals can't enter that city?

Write a program that implements the described system.

Input

The first line contains two integers N and E ($2 \leq N \leq 10^5$, $1 \leq E \leq 5 \times 10^5$), the number of cities and roads. Each of the following E lines contains two distinct integers between 1 and N – the labels of two cities connected by a road. There will be at most one road between any pair of cities.

The following line contains the integer Q ($1 \leq Q \leq 10^5$), the number of queries the system is being tested on. Each of the following Q lines contains either four or five integers. The first of these integers is the type of the query – 1 or 2.

If the query is of **type 1**, then the same line contains four more integers A , B , G_1 and G_2 as described earlier. A and B will be different. G_1 and G_2 will represent an existing road.

If the query is of **type 2**, then the same line contains three more integers A , B and C . A , B and C will be distinct integers.

The test data will be such that it is initially possible to get from each city to every other city.

Output

Output the answers to all Q queries, one per line. The answer to a query can be *da* (yes) or *ne* (no).

Example

Input	Output
13 15 1 2 2 3 3 5 2 4 4 6 2 6 1 4 1 7 7 8 7 9 7 10 8 11 8 12 9 12 12 13 5 1 5 13 1 2 1 6 2 1 4 1 13 6 7 8 2 13 6 7 2 13 6 8	da da da ne da